

VirtuBox Infotech Private Limited
Assessment Test -- Software Tester (QA)

Full Name	Gaurav Nagar
Email ID	gaurav.nagar.cs.2023@mitmeerut.ac.in
Contact Number	9548664986

Below is my response to the QA assessment for the Task Management Application -- test case design for Registration, Login, Task CRUD, and input validation, followed by a static bug/risk review.

1. Understanding & Approach

The app lets a user register, log in, and then create, view, edit, and delete tasks stored in a database. Since it's close to release with no existing test docs, I've approached this as black-box testing -- mapping each feature to its real user flow and covering it with positive, negative, and edge cases.

2. Test Case Design

2.1 Registration

TC ID	Scenario	Test Steps	Expected Result	Type
REG-01	Register with all valid details	Enter valid name, unique email, matching passwords; submit	Account created; redirected to login/dashboard with success message	Positive
REG-02	Register with an already registered email	Enter an email that already exists and submit	Rejected with "Email already registered" error	Negative
REG-03	Register with invalid email format	Enter email without '@' or domain and submit	Inline validation error; no server call made	Negative
REG-04	Register with a weak password	Enter a short/simple password (e.g. "123")	Blocked; password policy is displayed	Negative
REG-05	Password and Confirm Password do not match	Enter two different values in the two fields	Error shown; account not created	Negative
REG-06	Mandatory fields left blank	Leave Name/Email/Password empty, click Register	Required-field validation triggers; nothing sent to backend	Negative
REG-07	Leading/trailing or only-space Name	Enter " " or " Gaurav " in Name field	Whitespace-only rejected; extra spaces trimmed on save	Edge
REG-08	Maximum-length Name/Email input	Enter input at/just beyond the max character limit	Limit enforced gracefully; no silent truncation or crash	Edge
REG-09	Script/SQL-like input in Name field	Enter <script>alert(1)</script> or ' OR '1'='1	Input sanitized; no script runs, no SQL error exposed	Negative
REG-10	Double-click on Register	Click Register twice quickly on a valid form	Only one account created; duplicate ignored	Edge

2.2 Login

TC ID	Scenario	Test Steps	Expected Result	Type
LOG-01	Login with valid credentials	Enter correct email and password, click Login	Authenticated; redirected to dashboard	Positive
LOG-02	Login with incorrect password	Enter valid email with wrong password	Rejected with generic "Invalid email or password"	Negative
LOG-03	Login with unregistered email	Enter an email that doesn't exist	Same generic error (doesn't reveal if email exists)	Negative
LOG-04	Empty email or password field	Leave a field blank, click Login	Blocked client-side; no server call made	Negative
LOG-05	Different case in email	Register "User@Mail.com", login as "user@mail.com"	Case-insensitive match; login succeeds	Edge
LOG-06	Repeated failed login attempts	Enter wrong password 5+ times in a row	Temporary lock or CAPTCHA kicks in	Edge
LOG-07	Session persistence	Log in, then refresh or reopen the tab	Stays logged in until logout/token expiry	Positive
LOG-08	Direct access without login	Paste the dashboard URL while logged out	Redirected to login page	Negative

2.3 Task CRUD Operations

TC ID	Scenario	Test Steps	Expected Result	Type
TSK-01	Create a task with valid details	Fill required fields, click Save	Task saved and appears in the list instantly	Positive
TSK-02	Create a task with empty title	Leave Title blank, click Save	Validation error; task not created	Negative
TSK-03	View list after adding tasks	Create 3-4 tasks, open the task list	All tasks visible, correct details, no duplicates	Positive
TSK-04	Edit an existing task	Change title/description, click Update	Change saved and reflected instantly, no duplicate entry	Positive
TSK-05	Clear a mandatory field while editing	Delete Title text, click Update	Update blocked; original data preserved	Negative
TSK-06	Delete a task	Click Delete, confirm the action	Task removed from list and database	Positive
TSK-07	Cancel a delete	Click Delete, then Cancel on the prompt	Task remains unchanged	Edge
TSK-08	Edit/delete a task that's already gone	Open its edit/delete link after deleting it elsewhere	Graceful "Task not found" message, no crash	Edge
TSK-09	Very long title/description	Enter text far beyond expected length, save	Sensible limit enforced, or clean wrap/truncate	Edge
TSK-10	Access another user's task	As User A, try User B's task via URL/ID edit	Access denied -- users see only their own tasks	Negative

2.4 Input Validation & Error Handling

TC ID	Scenario	Test Steps	Expected Result	Type
VAL-01	Special characters/emoji in title	Enter symbols like @\$%&*() or emojis, save	Accepted and displayed correctly, or clearly restricted	Edge
VAL-02	Submit using Enter key	Fill a valid form, press Enter instead of clicking	Submits the same way, same validations apply	Positive
VAL-03	HTML/script tags in a text field	Enter <script>...</script> into Title/Description	Escaped and shown as plain text, never executed	Negative
VAL-04	Network failure while saving	Disconnect network during a Save/Update/Delete	Clear error/retry shown; no corrupted data saved	Edge
VAL-05	Alphabetic input in a numeric/date field	Enter letters into that field, try to save	Rejected before submission	Negative
VAL-06	Large pasted text in Description	Paste several paragraphs, save	Handled gracefully, or a clear limit applies	Edge
VAL-07	Rapid double-submission	Click Save twice quickly on a valid form	Only one record created	Edge
VAL-08	Error messages are field-specific	Trigger errors on Registration, Login, Task forms	Each error names the field and the issue clearly	Positive

3. Bug Identification (Static Review)

Based on requirement analysis alone (no execution), here are the potential bugs and risk areas I'd expect in this kind of registration/login/CRUD application:

#	Bug / Risk Area	Severity	Reason / Impact
1	No server-side validation (only client-side JS checks)	Critical	Bypassing the browser (e.g. via Postman) lets invalid or malicious data reach the database.
2	Passwords possibly unhashed or sent without HTTPS	Critical	A breach would directly expose user credentials.
3	Missing authorization on task edit/delete (IDOR)	Critical	One user could view or modify another user's tasks via ID/URL changes.
4	No account lockout or rate-limiting on login	Major	Login is open to brute-force password guessing.
5	Duplicate tasks on double-click / slow network	Major	No submit lock means the same task can be created twice.
6	No pagination for large task lists	Major	Performance degrades as the number of tasks grows.
7	Inconsistent error handling on failed API calls	Major	Silent failures leave users unsure whether an action worked.
8	Email uniqueness check may be case-sensitive	Minor	"User@mail.com" and "user@mail.com" could both register.
9	No confirmation before deleting a task	Minor	One accidental click could permanently remove a task.
10	Mandatory fields not visually marked	Minor	Users may resubmit incomplete forms without knowing why.

4. Summary of My Thought Process

- Mapped each feature to its real user flow first, then wrote test cases around that -- not the other way round.
- Balanced positive, negative, and edge cases for every feature, since most production bugs live at the edges.
- Picked bugs that are easy to miss in development but costly after release -- server-side validation, authorization, and duplicate submissions -- and ranked them by real impact.
- Kept everything traceable with unique TC IDs and feature-wise grouping so it's ready to hand off or drop into a test-management tool.